Program : **Diploma in Textile Technology**	
Course Code : **2069**	Course Title: **Basic Textile Mechanics Lab**
Semester : **2**	Credits: **No Credit**
Course Category: **Engineering Science**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To impart the basic knowledge of different mechanical elements used in Textile machines such as shaft, pulley, axle, key, coupling, bearing and their functions.
- To understand the concept of drive transmission and speed calculation.
- To study the application of different types of gears, belts, cam and tappet in textile machines.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I	1
Knowledge of basic Physics		Applied physics I	1
Knowledge of basic Physics		Applied physics II	2

Course Outcomes:

On completion of the course, the students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Identify the different mechanical elements used in Textile machines and explain their function.	9	Understanding
CO2	Illustrate and compare different methods of drive transmission	12	Understanding
CO3	Explain the different types of gears, their application and calculate speed from a drive transmission/gearing diagram.	13	Applying
CO4	Discuss the basic aspects of cam, tappet and variable speed drives.	8	Understanding
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	1						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	**Identify different mechanical elements used in Textile machines and explain their functions.**		
M1.01	Identification of various mechanical elements in textile machines.	4	Understanding
M1.02	Study of Shaft, axle, pulley, keys and coupling.	5	Understanding
CO2	**Illustrate and compare different methods of drive transmission**		
M2.01	Study of different Power Transmission Devices - Flexible drives and rigid drives used in textile machinery –comparison, merits and de merits.	4	Understanding
M2.02	Study of various belt drives, their suitability and function.	4	Understanding
M2.03	Study of roller chain drive – advantages and disadvantages	4	Understanding
	Lab Exam – I	1.5	
CO3	**Explain the different types of gears, their application and calculate speed from a drive transmission/gearing diagram.**		
M3.01	Study of different types of gear, identify the types of gears in the allotted machine and explain their application.	4	Understanding
M3.02	Study of worm & worm wheel drives and its types, pawl and ratchet drives	3	Understanding
M3.03	Study of gear train, drive transmission/gearing diagram and calculate the speed, belt slip%.	6	Applying

CO4	**Discuss the basic aspects of cam, tappet and variable speed drives.**		
M4.01	Study of Tappet and Cams.	4	Understanding
M4.02	Study of variable speed drives in Textile machines -step less cone pulleys , PIV drives	4	Understanding
	Lab Exam- II	1.5	

Text / Reference:

T/R	Book Title/Author
T1	Laboratory manual
R2	Ganapathy Nagarajan - Textile mechanisms for Spinning and Weaving machines
R3	R. Gokarneshan, Mechanics and calculations of Textile Machinery
R4	Slater K, Textile mechanics-vol 1, The Textile Institute
R5	Rengasamy R.S-Mechanics of Spinning machines-NCUTE

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in
2	www.sciencedirect.com./machine-design